

Role Research Documentation

Your name: Vxgx Vega

Role title: E-commerce Analyst

Reminder: Do not use AI to fill in this card. AI can be a thought partner to brainstorm, but the words should be your own.

1. Role Overview

Describe this role in your own words — what does this person do every day in a nutshell?

An E-commerce Analyst monitors, processes, and evaluates online sales data, website traffic, inventory velocity, and digital marketing performance. In a nutshell, this person acts as the analytical engine for an online brand—translating raw transactional and behavioral data into actionable operational decisions regarding pricing, ad spend, and stock management.

2. A Day in the Life

Walk through a realistic day for this person based on their key responsibilities. What are their major responsibilities? What takes the most time? What do they do that feels repetitive or manual?

A realistic day involves opening multiple disconnected dashboards (Shopify/WooCommerce, Amazon Seller Central, Google Analytics 4, Meta Ads Manager, and inventory ERPs).

- **Major Responsibilities:** Tracking daily revenue metrics, calculating customer acquisition costs (CAC) vs. return on ad spend (ROAS), monitoring inventory stockout risks, and compiling performance summaries for leadership.
- **What Takes the Most Time:** Manually exporting CSV files from different platforms, cleaning inconsistent column headers, removing duplicates, and stitching the data together inside complex master spreadsheets.
- **What Feels Repetitive or Manual:** Running daily manual data pulls to track sales performance by SKU, re-formatting raw CSV exports, and manually calculating inventory burn rates to flag reorder thresholds.

3. Top Pain Point

Describe the pain point this role experiences that you want to build a product solution for. Be specific, and draw from your role research and the email they sent.

The analyst suffers from severe **data processing lag caused by fragmented platform reporting**. Because sales, inventory, and ad metrics live in isolated silos, they are forced to spend over 60% of their working hours on manual data gymnastics—extracting and reformatting raw spreadsheets—rather than performing high-leverage predictive analysis. Consequently, stockouts occur on high-performing SKUs and ad spend is wasted on low-converting campaigns because insights are delivered days after trends emerge.

4. Root Cause Analysis

Pick the one pain point you want to build for. Then apply the 5 Whys — ask 'why?' five (or more) times to get past the surface complaint to the actual root cause. Fill in each level below.

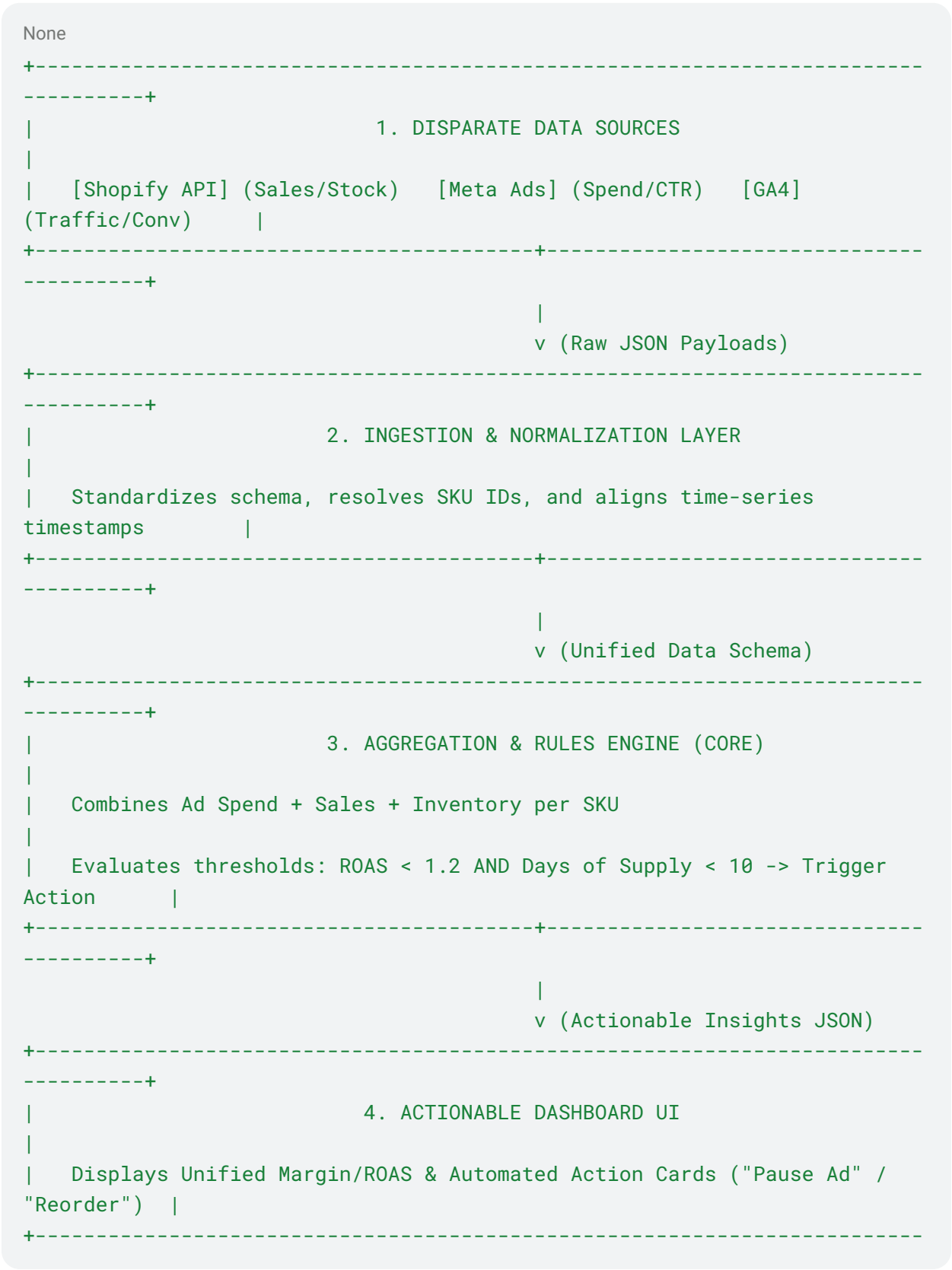
Why #1	<i>Why does this problem happen?</i> Because their daily performance reports take hours to compile, delivering stale insights after trends have already passed.
Why #2	<i>Why does that happen?</i> Because the analyst has to manually download, clean, and consolidate raw CSV files across 4+ different ad and sales platforms.
Why #3	<i>Why does that happen?</i> Because sales, marketing, and inventory metrics are stored in disconnected software silos that don't speak to one another.
Why #4	<i>Why does that happen?</i> Because there is no automated data ingestion or integration layer sitting between the sales platforms and the reporting dashboard.
Why #5	<i>This is the root cause — add more or less rows as necessary.</i> The brand lacks an automated data pipeline at the ingestion layer that normalizes and aggregates multi-platform metrics into a unified schema, forcing the analyst to act as a human data transformer.

5. Problem Statement

Write the problem statement for this role's pain point. Be as specific as possible — the more specific you are, the clearer your product will be. Format: [Role] struggles with [specific pain point] because [root cause], which means [consequence].

An **E-commerce Analyst** struggles with **stale reporting and delayed decision-making regarding inventory and ad spend** because **multi-platform sales and marketing data are un-indexed and fragmented at the ingestion layer**, which means **the brand wastes capital on underperforming ad campaigns, suffers preventable stockouts on top SKUs, and loses thousands in lost revenue**.

MVP Solution Logic Diagram



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System Breakdown & Connectivity

1. **Disparate Data Sources (Inputs):** Mock payload streams representing sales/inventory data (Shopify) and campaign performance metrics (Meta Ads & GA4).
2. **Ingestion & Normalization Layer:** Cleans, maps, and standardizes disparate multi-platform schemas into a single unified JSON structure anchored by **SKU_ID** and timestamp.
3. **Aggregation & Rules Engine (THE ABSOLUTE CORE):** The central brain of the product. It joins ad spend against real-time revenue and stock levels to evaluate business logic (e.g., calculating true contribution margin and flagging campaigns driving sales to out-of-stock SKUs).
4. **Actionable Dashboard UI (Output):** Converts the processed engine output into a streamlined executive view featuring real-time KPI metrics and immediate action cards.

The mock JSON data schema and ingestion script joining Shopify and Meta Ads. Before touching any dashboard UI, I need to define and populate the unified data structure. Tomorrow's priority is creating mock payloads for both APIs—Shopify_Inventory_Payload (SKU, Stock Level, Unit Cost, Retail Price) and Meta_Ads_Payload (Campaign ID, Target SKU, Spend, Impressions, Conversions)—and building the transformation function that normalizes them into a single SKU_Performance_Object keyed by SKU_ID. Once that schema works and outputs a unified JSON array, the core engine has real data to run rules against—making the UI build later in the week straightforward.

Time to build. Focus on your one core feature first — the thing your product must do. Come back to your Product Card if scope creep starts.

If you get stuck:

Paste your error message directly and I'll help debug it

Describe what you're trying to do and what's not working — be specific

Ask 'why does this look wrong' and describe what you're seeing

Your goal is to Complete the task objectives for MVP Build Session I.

Question 1: What is the one core feature you're building today, and what does "working" look like for it?